

3. PRINCIPLE OF VIRTUAL WORK

Principle of virtual work states that the algebraic sum of the virtual work done by external active forces on an ideal mechanical system is zero for any and all virtual displacements consistent with the constraints.

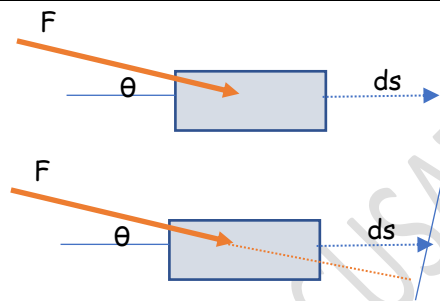
Work of a force

Consider a constant force F acting on the body which moves a distance ds . By definition the work done by the force F on the body

$$U = (F \cos \theta) ds$$

This will be also equal to

$$U = F (ds \cos \theta)$$



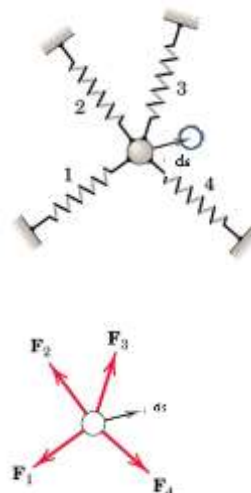
Any assumed and arbitrary small displacement from the equilibrium position consistent with the system constraints is known as **virtual displacement**. The term **virtual** is used to indicate that the displacement does not really exist but only is assumed to exist so that we may compare various equilibrium positions to obtain the correct one.

Consider a small body connected by four springs as shown. The forces acting on the body are F_1, F_2, F_3, F_4 and the body is in equilibrium.

$$\Sigma F_x = 0; \quad \Sigma F_y = 0, \text{ where } x-y \text{ are two mutually perpendicular axes.}$$

Let this body be given a small virtual displacement ds as shown. If $\theta_1, \theta_2, \theta_3, \theta_4$ are the angles that the forces make with displacement direction. The virtual work done by the forces over this virtual displacement is

$$\begin{aligned} U &= F_1 \cos \theta_1 ds + F_2 \cos \theta_2 ds + F_3 \cos \theta_3 ds + F_4 \cos \theta_4 ds \\ &= (F_1 \cos \theta_1 + F_2 \cos \theta_2 + F_3 \cos \theta_3 + F_4 \cos \theta_4) ds \\ &= (\Sigma F_x) ds = 0 \text{ (Assuming the direction of displacement as } x\text{-axis)} \end{aligned}$$

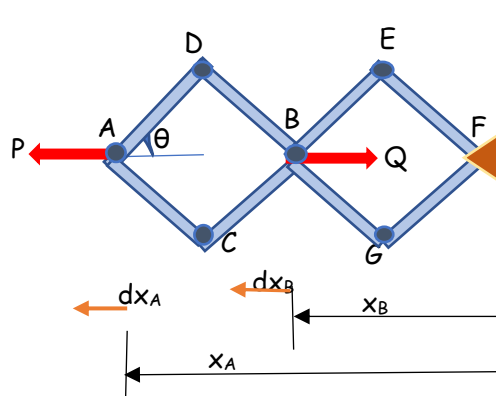


One advantage of this method is that we only need to consider the external forces. Though work is a scalar quantity, it is considered negative if the force and displacement are in opposite directions.

Q 1. What is the relation between the forces P and Q for the equilibrium of the frame consisting of two identical rhombuses with hinged joints?

Let the side of the rhombus be ' a '.

x_A be the position of joint A and x_B the position of joint B from the fixed wall where A and B are the points where forces P and Q act.



$$x_A = 4a \cos \theta \quad \text{and} \quad x_B = 2a \cos \theta$$

Let us give a very small virtual displacement to the system. As per the system configuration, there exist two possibilities. We can move the end A either towards right or towards left, or we can

rotate end A through a very small angle about the end F. For very small rotation about F, the displacements of points A and B will be perpendicular to the forces. Hence the virtual work done by these forces will be invariably zero. Therefore, we can consider a small virtual displacement to point A towards right.

Then the differential change in the position of A, $dx_A = -4a \sin \theta d\theta$

The differential change in the position of B, $dx_B = -2a \sin \theta d\theta$

The negative sign shows that x_A and x_B are inversely related to θ . That is; when θ increases, both x_A and x_B decreases and vice versa.

The magnitude of the virtual displacements $dx_A = 4a \sin \theta d\theta$ and $dx_B = 2a \sin \theta d\theta$.

According to the Principle of Virtual Work, in an equilibrium system the algebraic sum of virtual work done by the external forces over a virtual displacement is zero.

Virtual work done by P = $P dx_A = P 4a \sin \theta d\theta$ (work is positive as P and dx_A have same direction)

Virtual work done by Q = $-Q dx_B = -Q 2a \sin \theta d\theta$ (work is -ve as Q and dx_B are opposite)

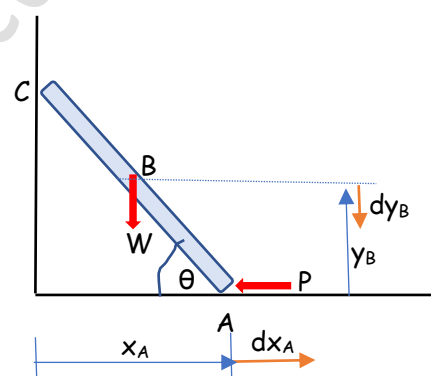
For equilibrium, the total virtual work done by active forces should be zero.

$$P 4a \sin \theta d\theta - Q 2a \sin \theta d\theta = 0$$

$$P = \frac{Q}{2}$$

Q 2. What is the magnitude of force P required to keep the bar of weight W and length l in equilibrium, if the floor and wall are smooth?

Let us give a small virtual displacement to the system. The end A can be moved towards left or right. Then the position of points A and B (the points of applications of external forces P and Q) will change. We need the change in position in the line of the forces.



Let x_A be the position of point A from the fixed wall and y_B the position of point B from the floor.

$$x_A = l \cos \theta \quad \text{and} \quad y_B = \frac{l}{2} \sin \theta$$

Assume that the end A is given a virtual displacement towards right.

Then the differential change in the position of A, $dx_A = -l \sin \theta d\theta$

The negative sign shows that x_A is inversely related to θ . That is; when θ increases, x_A decreases and vice versa.

The differential change in the vertical position of B, $dy_B = \frac{l}{2} \cos \theta d\theta$

The magnitude of the virtual displacements $dx_A = l \sin \theta d\theta$ and $dy_B = \frac{l}{2} \cos \theta d\theta$.

According to the Principle of Virtual Work, in an equilibrium system the algebraic sum of virtual work done by the external forces over a virtual displacement is zero.

Virtual work done by P = $-P dx_A = -P l \sin \theta d\theta$ (work is -ve as P and dx_A are opposite)

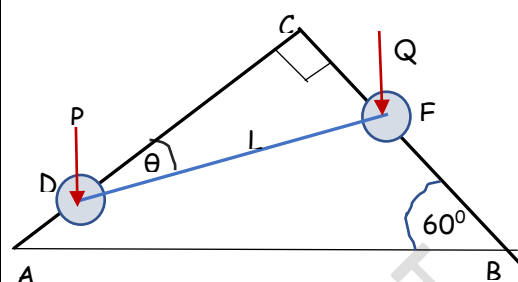
Virtual work done by W = $W dy_B = W \frac{l}{2} \cos \theta d\theta$ (work is +ve as W and dy_B are in the same direction)

For equilibrium, the total virtual work done by active forces should be zero.

$$= -P l \sin \theta d\theta + W \frac{l}{2} \cos \theta d\theta = 0$$

$$P = \frac{W}{2} \cot \theta$$

Q 3. Using the principle of virtual work, find the value of angle θ defining the equilibrium configuration of the system. The balls D and E can slide freely along the smooth bars AC and BC. The string DE connecting the balls is inextensible.



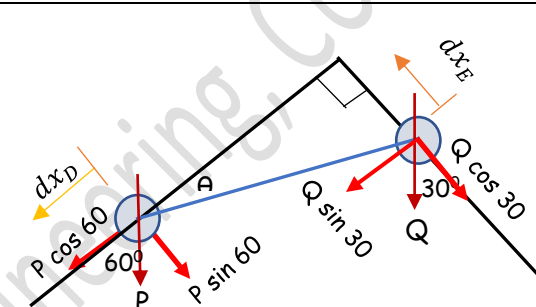
Let us assume that the ball D is given a virtual displacement downward along the bar AC. Since the string is inextensible, the ball E will move up. The positions of both the balls D and E will change.

The position of D from the corner C, $x_D = L \cos \theta$

The position of E from the corner C, $x_E = L \sin \theta$

The differential change in the position of D, $dx_D = -L \sin \theta d\theta$

The negative sign shows that x_D is inversely related to θ . That is; when θ increases, x_D decreases and vice versa. The differential change in the position of E, $dx_E = L \cos \theta d\theta$



The magnitudes of the virtual displacements, $dx_D = L \sin \theta d\theta$, $dx_E = L \cos \theta d\theta$

The virtual work done by the active external force P = $+P \cos 60 dx_D = +P \cos 60 L \sin \theta d\theta$

The virtual work done by the active external force Q = $-Q \cos 30 dx_E = -Q \cos 30 L \cos \theta d\theta$

For equilibrium, the total virtual work done should be zero.

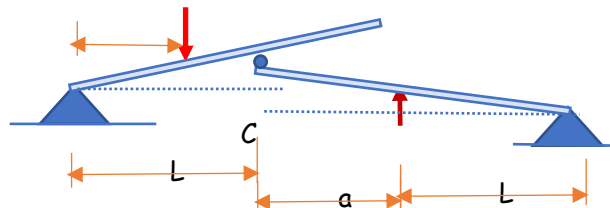
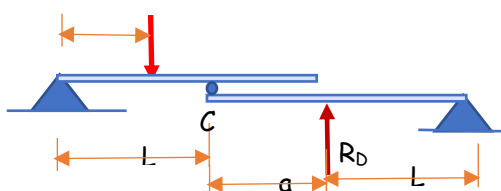
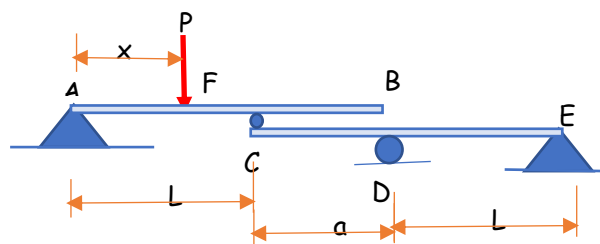
$$+P \cos 60 L \sin \theta d\theta - Q \cos 30 L \cos \theta d\theta = 0$$

$$\tan \theta = \frac{\sqrt{3} Q}{P}$$

So far, we have been considering the virtual work done by active external forces. But we can extend this technique to reactive forces also.

Q 4. Using the principle of virtual work, calculate the reaction at the roller support at D.

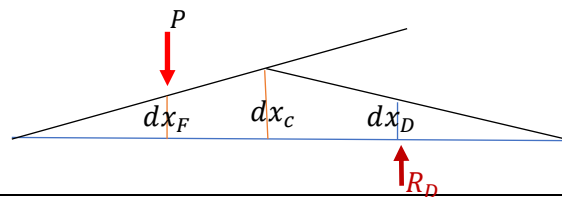
The support at D is replaced by the reaction R_D . The end C is given a small virtual displacement dx_C upwards. Then the points of applications of the forces P and R_D will also move.



From the similar triangles, we can write

$$\frac{dx_c}{L} = \frac{dx_F}{x} \Rightarrow dx_F = \frac{x}{L} dx_c$$

$$\frac{dx_c}{L+a} = \frac{dx_D}{L} \Rightarrow dx_D = \frac{L}{L+a} dx_c$$



Virtual work done by $P = -P dx_F = -P \frac{x}{L} dx_c$

Virtual work done by $R_D = +R_D dx_D = +R_D \frac{L}{L+a} dx_c$

By the principle of virtual work, the total virtual work done is zero if the system is in equilibrium.

$$-P \frac{x}{L} dx_c + R_D \frac{L}{L+a} dx_c = 0$$

$$R_D = \frac{(L+a)x}{L^2} P$$

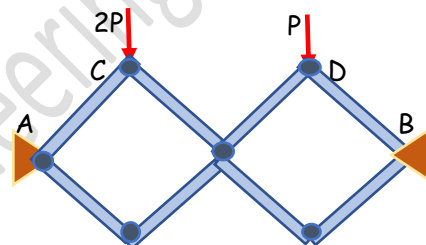
Q 5. Using the principle of virtual work, calculate the reaction at B. The bars form two identical squares.

The support at B is replaced by the reaction components X_B and Y_B

Let the side of the square be 'a'.

The position of B from the support at A,

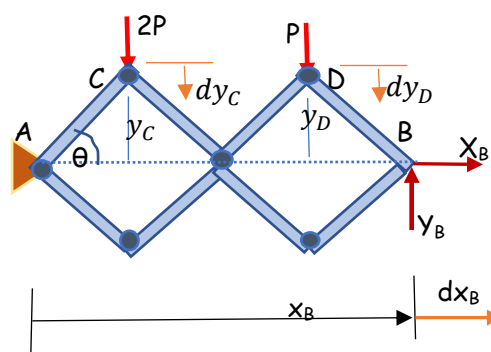
$$x_B = 4a \cos \theta$$



Though we know that $\theta = 45^\circ$, we will keep it as θ for the time being and substitute its value at the end.

Let us now examine the system and identify the possible virtual displacements. We can move the end B towards right or left, or we can rotate the frame with respect to the hinge at A. Let us assume that the end B is given a small virtual displacement towards right.

The differential change in the position of B,
 $dx_B = -4a \sin \theta d\theta$



The magnitude of $dx_B = 4a \sin \theta d\theta$

When end B is moved towards right, the points C and D will move down. The vertical distance of C and D will change. The vertical distance of C and D from the reference line AB is

$$y_C = y_D = a \sin \theta$$

The differential change in the vertical positions of C and D,

$$dy_C = dy_D = a \cos \theta d\theta$$

The total virtual work done by the forces = $+X_B dx_B + 2P dy_C + P dy_D$ (The force Y_B will not do any work as the force is perpendicular to the displacement dx_B)

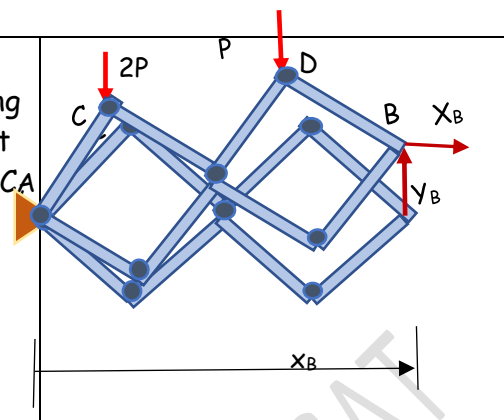
By the principle of virtual work, if a system of forces is in equilibrium the total virtual work done by the forces is zero

$$+X_B 4a \sin \theta d\theta + 2Pa \cos \theta d\theta + P a \cos \theta d\theta = 0$$

$$X_B = -\frac{3Pa \cos \theta}{4a \sin \theta} = -\frac{3P \cot 45}{4} = -\frac{3P}{4}$$

Now let us give a small virtual rotation $d\theta$ to the frame about the hinge at A. Then the entire frame will be rotating about A through the angle $d\theta$. The point B will rotate about A with radius AB, point C will rotate about A with radius AC and point D will rotate about A with radius AD. Since the rotation is very small, the displacements z_B, z_C, z_D will be perpendicular to the respective radii AB, AC, and AD.

$$AB = 4a \cos \theta = 2\sqrt{2} a, \quad AC = a$$



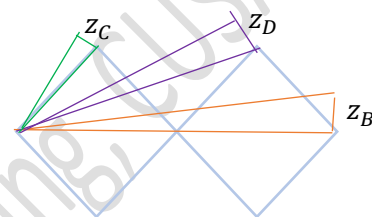
$$AD^2 = \left(\sqrt{2} a + \frac{a}{\sqrt{2}}\right)^2 + \left(\frac{a}{\sqrt{2}}\right)^2 = \frac{9a^2}{2} + \frac{a^2}{2} = \frac{10a^2}{2} = 5a^2$$

$$AD = \sqrt{5} a$$

$$z_B = AB \cdot d\theta = 2\sqrt{2} a d\theta$$

$$z_D = AD \cdot d\theta = \sqrt{5} a d\theta$$

$$z_C = a d\theta$$



Now for finding the virtual work done by the forces, we need to resolve the forces in the line of displacements or displacements in the line of forces.

Here, Y_B is along and X_B is perpendicular to the line of z_B

The virtual work done by $Y_B = +Y_B \cdot z_B = +Y_B 2\sqrt{2} a d\theta$ (X_B will not do work)

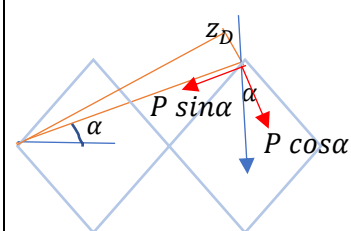
The force at D is resolved parallel and perpendicular to z_D (or AD).

The angle that AD makes with horizontal $\tan \alpha = \frac{\frac{a}{\sqrt{2}}}{\frac{3a}{\sqrt{2}}} = \frac{1}{3} \rightarrow$

$$\alpha = 18.4^\circ$$

The component of P at D along the line $z_D = P \cos 18.4 = 0.95 P$

The virtual work done by $P \cos 18.4 = -P \cos 18.4 z_D = -0.95 P \sqrt{5} a d\theta$



The perpendicular component $P \sin 18.4$ will not do work for the displacement z_D

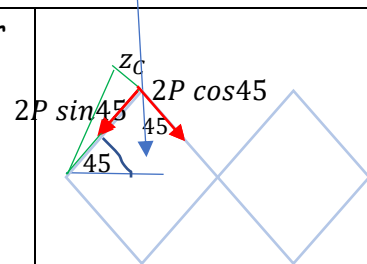
The force at C is resolved into components along and perpendicular to z_C (or AC).

The component of $2P$ along $z_C = 2P \cos 45$

The virtual work done by this force

$$= -2P \cos 45 z_C = -2P \cos 45 a d\theta$$

The perpendicular component $P \sin 45$ will not do any work for this virtual displacement.



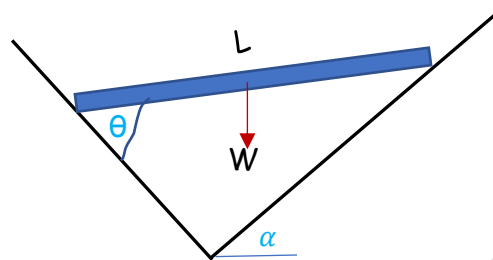
For equilibrium, the algebraic sum of the virtual work done by all forces must be zero. Hence,

$$+Y_B 2\sqrt{2} a d\theta - 0.95 P \sqrt{5} a d\theta - 2P \cos 45 a d\theta = 0$$

$$Y_B = \frac{0.95\sqrt{5} + 2 \cos 45}{2\sqrt{2}} P = 1.25 P$$

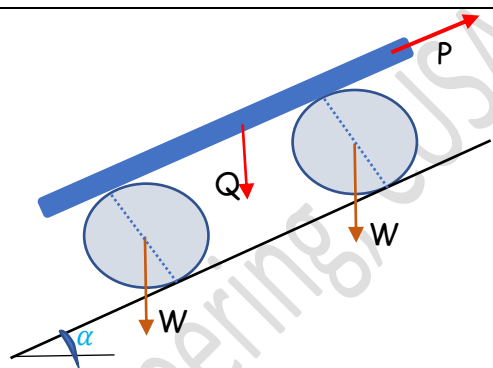
#Exercise 1

A rigid bar AB is supported in a vertical plane by mutually perpendicular smooth surfaces OA and OB as shown. Using the principle of virtual work, calculate the angle θ defining the configuration of equilibrium.



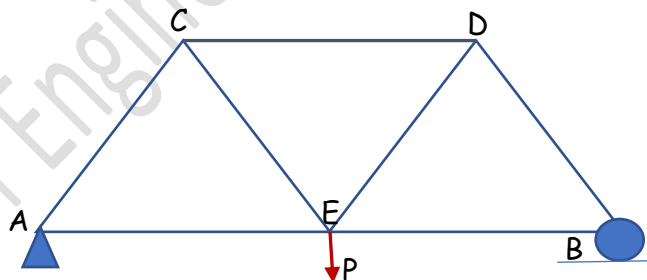
#Exercise 2

A beam supported on two rollers, each of radius 'r' is moved up an inclined plane by a force P which acts parallel to the plane. Determine the magnitude P necessary to maintain equilibrium if there is no slipping between rollers and the inclined plane or the beam and the rollers. The weight of the beam is Q and the weight of each roller is W.



#Exercise 3

Using the principle of virtual work, find the force in the bar CD of the truss. Each bar is of length L.



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